

Functional Requirements & Software Requirements Specification

E-commerce & Campaign Profitability Analyzer

1. Document Overview

1.1 Purpose

The E-commerce & Campaign Profitability Analyzer provides a reconciled view of e-commerce revenue, costs, campaign performance, payments, refunds, and customer value - closing the gap between marketing-level and finance-level reporting.

1.2 Scope

In Scope
Customer and order management
Product and order-item economics
Payment and refund reconciliation
Advertising campaign performance
Revenue and profitability calculations
Customer value and dormancy analysis
SQL-based business analysis and Power BI reporting

1.3 High-Level Users

User	Primary Requirement
Finance	Revenue, COGS, profit and reconciliation
Marketing	Campaign performance and ROAS
Management	Business performance and trends
Business Analyst	Analysis, validation and reporting

2. Functional Requirements

FR-1.1 - Cancelled Order Treatment

Definition	Cancelled orders do not contribute to qualifying revenue, COGS, contribution profit, or customer lifetime spend.
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FR-1.2 - Net Revenue

Definition	Qualifying product revenue after applicable discounts and refunds.
Formula	Net Revenue = Gross Product Revenue – Discounts – Refunds

FR-1.3 - Cost of Goods Sold

Definition	Calculated from product quantity and unit cost.
Formula	$COGS = \sum(Quantity \times Unit\ Cost)$

FR-1.4 - Contribution Profit

Definition	Net Revenue after deducting COGS and allocated advertising spend.
Formula	Contribution Profit = Net Revenue – COGS – Allocated Ad Spend

FR-1.5 - Campaign Analysis

Definition	Campaign-level revenue, ad spend, ROAS, and contribution profitability. ROAS returns NULL when ad spend is zero.
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FR-1.6 - Campaign Ranking

Definition	Campaign/channel performance ranked using DENSE_RANK() on the selected profitability metric.
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FR-1.7 - Customer Segmentation

Definition	Customers classified by qualifying completed spend. VIP threshold: $\geq \$1,500$ lifetime qualifying spend.
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FR-1.8 - Customer Dormancy

Definition	Customers with 60+ days since their latest qualifying completed order are flagged dormant.
Formula	Days Since Last Order = Current Date – Latest Qualifying Order Date

FR-1.9 - Payment & Refund Reconciliation

Definition	Order-level expected amounts are compared against recorded payments to identify discrepancies.
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FR-1.10 - Revenue Growth

Definition	Month-over-month revenue growth using the previous month's qualifying revenue via LAG().
Formula	MoM Growth % = $(\text{Current Month Revenue} - \text{Previous Month Revenue}) \div \text{Previous Month Revenue} \times 100$

3. Data & System Requirements

3.1 Core Data Entities

#	Table
1	Customers
2	Ad_Campaigns
3	Products
4	Orders
5	Order_Items
6	Payments
7	Refunds

3.2 Relationships

- Customers 1 → Many Orders
- Ad_Campaigns 1 → Many Orders
- Orders 1 → Many Order_Items
- Products 1 → Many Order_Items
- Orders 1 → Many Payments
- Orders 1 → Many Refunds

3.3 Table Purpose

Table	Purpose
Customers	Customer identification and info required for lifetime value and dormancy analysis
Ad_Campaigns	Campaign info and ad spend required for profitability and ROAS
Products	Product info and unit cost required for product economics and COGS
Orders	Order-level transactions including customer, campaign, status, financials
Order_Items	Products purchased, quantities, and selling prices per order
Payments	Payment transactions associated with orders
Refunds	Refund transactions associated with orders

3.4 Data Integrity

- Valid primary and foreign-key relationships
- Valid order-to-customer and order-to-campaign relationships
- Valid product references in order items
- Valid payment and refund references
- Consistent financial calculations

4. Business Logic & Calculations

Metric	Formula
Gross Product Revenue	$\Sigma(\text{Quantity} \times \text{Unit Price})$
Discount Amount	Sum of applicable order discounts
Net Revenue	Gross Revenue – Discounts – Refunds
COGS	$\Sigma(\text{Quantity} \times \text{Unit Cost})$
Contribution Profit	Net Revenue – COGS – Allocated Ad Spend
ROAS	Qualifying Revenue $\div$ Ad Spend (NULL if Ad Spend = 0)
Customer Lifetime Spend	Sum of qualifying completed customer spend
Customer Dormancy	Flagged at 60+ days since latest qualifying order
MoM Growth	Current vs. preceding month revenue, via LAG()

5. System Behaviour & SQL Requirements

5.1 SQL Analysis Requirements

The system supports SQL analysis involving:

- Joins, aggregations, and conditional calculations
- CTEs and window functions
- Ranking and date-based analysis
- Customer segmentation and reconciliation queries

5.2 Null & Zero Handling

- Returns NULL where a ratio cannot be meaningfully calculated
- Avoids division-by-zero errors
- Handles missing optional transaction data appropriately
- Prevents cancelled orders from being treated as qualifying revenue

5.3 Data Validation

- Financial calculations are consistent
- Foreign-key relationships are valid
- Cancelled orders are excluded where required
- Refunds are correctly reflected
- Customer segmentation follows defined thresholds

6. Reporting & Analytical Outputs

Category	Outputs
Financial Performance	Gross Revenue, Discounts, Refunds, Net Revenue, COGS, Contribution Profit
Campaign Performance	Ad Spend, Revenue, ROAS, Contribution Profit, Campaign ranking
Customer Analysis	Lifetime Spend, VIP status, Days Since Last Order, Dormancy
Revenue Trends	Monthly Revenue, Previous Month Revenue, MoM Growth %
Reconciliation	Order amount, Payment amount, Refund amount, Discrepant amounts

7. Traceability & Acceptance Summary

7.1 Business Requirement Traceability

Business Requirement	Related Functional Requirement
BR-1 Reconciled Profitability	FR-1.1 – FR-1.4
BR-2 Campaign & Channel Profitability	FR-1.5 – FR-1.6
BR-3 VIP Customer Retention	FR-1.7 – FR-1.8

Business Requirement	Related Functional Requirement
BR-4 Revenue Growth	FR-1.10
BR-5 Cash & Refund Reconciliation	FR-1.9
BR-6 Product-Level Economics [Not yet implemented]	FR-1.3 – FR-1.4 (partial)

7.2 Acceptance Criteria

The system is functionally acceptable when:

- Cancelled orders are excluded from qualifying financial metrics
- COGS correctly reflects product unit cost and quantity
- Contribution profit deducts COGS and applicable ad spend
- VIP customers are correctly identified using the defined threshold
- Dormant customers are correctly identified using the 60-day rule
- Payment and refund discrepancies can be identified
- MoM revenue growth correctly compares consecutive months
- Campaign performance and ROAS handle zero ad spend correctly

This document provides the implementation baseline for the database, SQL analysis, and Power BI reporting components of the project. | Team: Shreyans Jain & Aadhya Patel